AI-use disclosure (outside report page limit)

Team: Prakhar Keshniya, Harsh Sahu, Pranav Keshniya · Code, outputs and run instructions: https://github.com/Harshu3605/quantum-traffic-gridlock-qaoa

AI tool used. Claude (Anthropic, model Claude Opus 5.5) through Claude Code, a coding assistant. The rules (Sec. 10) permit AI assistants and general-purpose libraries, provided teams declare the assistance and verify the results independently. This page does that.

Who did what.

Area	Team (humans)	AI assistant
Problem understanding and scope	Read the problem statement and set the goal: an end-to-end pipeline that follows every PS requirement and reports results honestly	Turned the PS into a requirement checklist (CLI, files, limits, baselines, metrics)
Official inputs	Obtained the scorer and scenario R. Followed up with the organisers until scenario H was released, then supplied H	Used the files unmodified and ran H with unchanged code
Modelling and QUBO	Required strict compliance with the PS rules, including no hard-coded or answer-encoding QUBO, and accepted the documented rule	Proposed and implemented the per-approach decomposition and the order-2 Walsh projection
QAOA and experiments	Required compliance with the PS resource limits and honest reporting of results	Wrote the QAOA code, the baselines, the ablations and the 30-scenario generalisation test
Submission checks	Caught that a pasted submission checklist belonged to another track, and confirmed the correct one	Mapped every item of the correct checklist to a file
Repository and submission	Chose the repository name, created the public GitHub repo, made the initial commit and push, and uploads the Drive submission	Added later commits and the `v1.x-submission` tags, and assembled the Drive folder and zip
Documentation	Asked for a plain-language explainer so the team could understand and present the work, and reviewed the outputs	Drafted the README, the report, this disclosure and the explainer

How the results were verified (not taken on trust from the AI).

- Every reported cost comes from the organisers' `official/scorer.py`, used unmodified. It reproduces the PS verification costs 116, 146, 237 and 92.5.
- Our simulator and the QUBO decomposition were asserted equal to the official scorer on 40 random scenarios × 256 schedules. The QUBO↔Ising conversion was also checked (52 automated tests).
- The Qiskit circuit and the trained numpy state agree to a TVD of about 1e−15. A test re-imports `circuit.qasm`.
- Results are deterministic per seed. Anyone can regenerate every number with the commands in README.md.

Responsibility. The team takes responsibility for the submission, including the AI-assisted parts.